Jacob Su

jacob.yuanfeng.su@gmail.com | 740.604.0196 | Plain City, OH
github.com/pekingperson | linkedin.com/in/jacob-su-a13038284

Education

The Ohio State University, College of Engineering Columbus, OH
B.S. in Computer Science and Engineering, Minor in Robotics Expected May 2028
GPA: 3.767 | Degree in progress | Fundamentals of Engineering Honors (FEH)
Coursework: Software I, Software II, Linear Algebra, Calculus III

Technical Skills

Languages: Python, C++, Java, TypeScript, MATLAB
Software: Next.js, FastAPI, Docker, Git/GitHub, JWT authentication, Chrome/Edge extensions
AI/ML and Robotics: TensorFlow, vision transformers, parameter-efficient fine-tuning, Isaac Sim, Ubuntu Linux, Weights & Biases, robot calibration, teleoperation, CAD/Onshape/SolidWorks

Software Projects

Kinematics Hub | Next.js, FastAPI, Docker, JWT

- Developed a Dockerized robotics web platform with a Next.js frontend, FastAPI backend, JWT authentication, and modular services.

Recommendation Memory | TypeScript, Chrome/Edge

- Built a browser extension that locally stores and searches 20 YouTube recommendation sessions.

Additional Software and Machine Learning Projects

- Built a speech-to-IPA transcription program and a TensorFlow model for cartographic projection generation.
- Developed a C++ video game through the FEH program; built an interactive geographic mapping site integrating Census Reporter, Census Dots, and Opportunity Atlas data.

Experience

Elorian.AI / AxioForge AI Partnership Remote
Volunteer AI Systems Intern / AxioForge Founder May 2026 - Present

- Implement applied AI systems and prompt-to-CAD workflows for visual reasoning and engineering use cases in partnership with Elorian.AI.
- Develop CAD model-generation benchmarks; automate builds, evaluate outputs and failure modes, and document reproducible results.

Ohio State University Machine Learning as a Basis Lab Columbus, OH
Intern Oct 2025 - Present

- Worked on vision transformer (ViT) research studying parameter-efficient fine-tuning (PEFT) for visual recognition.
- Supported CLARE-related robotics research in continual learning for vision-language-action models and adapter-based learning.
- Helped transfer simulation workflows to real-robot testing, connecting model evaluation with physical robot operation.

DISL (Design, Innovation, and Simulation Laboratory), Scott Laboratory Columbus, OH
Intern Jul 2024 - Aug 2024

- Calibrated a Universal Manipulation Interface (UMI) robot and gripper for reliable manipulation trials.
- Set up the UMI interface using Python and Isaac Sim on Ubuntu for trial runs and simulation-based testing.
- Recorded hundreds of robot-operation videos used to train an AI model for manipulation research.

Honors and Additional Skills

National Merit Finalist | Ohio State Dean's List | Fluent Mandarin Chinese